

LIQUIDITY TWIN

Regulatory Liquidity Digital Twin, Control Graph and Reporting Compiler

Institutional Operational Specification & Executive Report

A controlled regulatory simulation modeling balance-sheet mutation, double-entry accounting reconciliation, Basel III classification rules (BCBS 295 / 238), continuous automated controls, exact Shapley ratio movement decomposition, bidirectional lineage provenance, and publication-ready reporting.

PROJECT DELIVERABLES	LOCATION / ACCESS LINK
Analyst Workstation Dashboard 11 views, interactive ECharts waterfall, React Flow DAG	Open Local Workstation (http://localhost:5173)
Project Report (Compiled PDF) Multi-page publication document with figures and sign-offs	reports/generated/Liquidity_Twin_Project_Report.pdf
Regulatory Reporting Pack (PDF & XLSX) Formal 2-page schedule summary and 3-sheet formulaic model	reports/generated/Liquidity_Reporting_Pack_2026-Q3.pdf reports/generated/Liquidity_Reporting_Pack_2026-Q3.xlsx
Source Code Repository Verified implementation, 229 automated tests, CI automation	github.com/hriday-sobti/liquidity-twin-regulatory-control

Reported Metric	Baseline Value	Supervisory Standard	Regulatory Floor	Compliance Surplus
Net Stable Funding Ratio (NSFR)	117.64%	BCBS 295 (Oct 2014)	100.00%	+17.64 pp (Compliant)
Liquidity Coverage Ratio (LCR)	132.41%	BCBS 238 (Jan 2013)	100.00%	+32.41 pp (Compliant)
Available Stable Funding (ASF)	\$842.30M	BCBS 295 §§17–31	N/A	\$842,300,000.00 Capital/Liab.
Required Stable Funding (RSF)	\$716.00M	BCBS 295 §§32–45	N/A	\$716,000,005.00 Assets/OBS
HQLA Liquidity Buffer	\$219.00M	BCBS 238 §§49–64	N/A	Level 1 (\$185M) + 2A (\$34M)
30-Day Net Cash Outflow	\$165.30M	BCBS 238 §§67–153	N/A	Gross (\$167.1M) - Inflows (\$1.8M)
Balance Sheet Parity Variance	\$0.00	Double-Entry Identity	\$0.00	Assets (\$1.05B) == L+E (\$1.05B)
Continuous Control Mesh Health	98.0%	BCBS 239 Standards	>=95.0%	98 Pass / 1 Warning / 1 Break
Automated Test Verification	229 / 229	Pytest Test Harness	100.0%	100% Passing in ~4.3s local run

Model Institution: Synthetic Benchmark Bank NA (Entity: ENT-001) | **Close Cycle:** CYCLE-2026-Q3 | **Snapshot:**

SNAP-2026-Q3-BASE

Author / Lead Analyst: Hriday Singh Sobti | **Compiled:** 2026-10-02 18:29:29 UTC | **Classification:** Internal Institutional Risk & Control Documentation

1. Executive Summary & Problem Context

Institutional treasury, risk, and financial control departments operate under intense supervisory scrutiny. Following the Basel III financial regulatory reforms, banking institutions must continuously satisfy and report two structural liquidity constraints: the **Net Stable Funding Ratio (NSFR)** across a one-year funding horizon, and the **Liquidity Coverage Ratio (LCR)** across an acute 30-day stressed outflow horizon.

While calculating these ratios on a spreadsheet is mathematically trivial, real-world reporting pipelines suffer from four severe operational limitations: (1) **Black-Box Variance**: When ratios shift between close periods, teams cannot readily decompose the movement into underlying asset and liability drivers. (2) **Reconciliation Breaks**: Balance sheets often fall out of parity post-freeze due to unrecorded late adjustments. (3) **Lineage Opacity**: Tracing a published metric back to raw source transactions (the 'Metric Birth Certificate') requires hours of ad-hoc database querying. (4) **Stress Blindness**: Treasury cannot easily simulate the impact of market deposit runs without cloning or risking production datasets.

Liquidity Twin is engineered to eliminate these friction points by providing a fully event-driven, auditable digital twin that connects transaction ingestion, double-entry validation, rule-based classification, continuous control execution, movement attribution, stress testing, and publication packs into one reproducible workflow.

2. Newly Contributed Architectural Capabilities

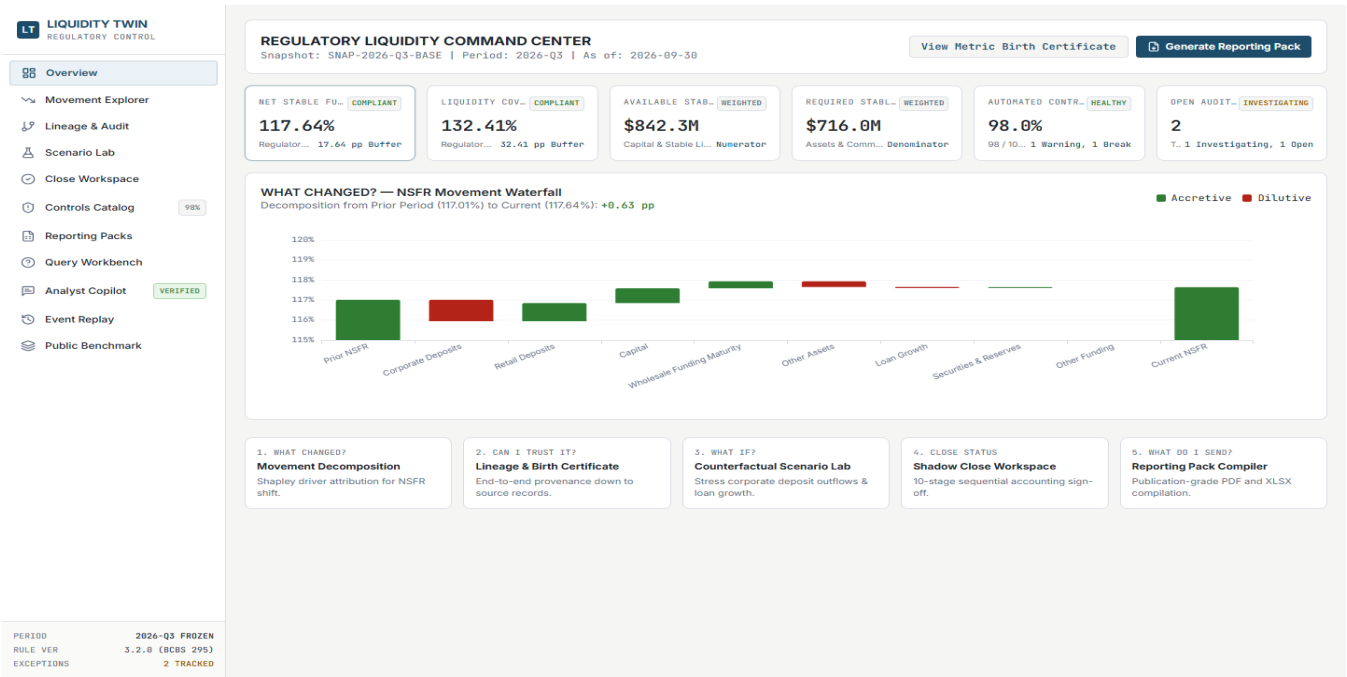
The system goes beyond an academic calculator by introducing ten distinct production-grade contributions:

- **Event-Driven Ingestion Foundation**: Processes 1,014 append-only events across 5 currencies into 20 funded positions, enabling transaction-level forensic inspection.
- **Automated Double-Entry Parity Guard**: Enforces $\$Assets == Liabilities + Equity\$$ with exact \$0.00 variance before any regulatory calculation is permitted.
- **Versioned Regulatory Rule Engine**: Implements 26 centralized, versioned classification rules explicitly citing BCBS 295 and 238 standards.
- **Exact Additive Shapley Movement Decomposition**: Decomposes ratio shifts into underlying funding drivers with zero residual error (exact additive reconciliation).
- **Directed Acyclic Graph Lineage (86 Nodes, 88 Edges)**: Enables three-click backward provenance traversal ('Metric Birth Certificate') and forward control blast-radius impact analysis.
- **100 Continuous Automated Controls**: Executes checks across 10 operational families on every close cycle (98.0% baseline pass rate, modeling real-world operational health).
- **Structured Exception Lifecycle**: Manages reconciliation breaks with exposure estimation and formal audit status tracking (OPEN -> INVESTIGATING -> REMEDIATED -> CLOSED).
- **Sequential 10-Stage Shadow Close Workflow**: Implements formal operational milestones with automated hard gates blocking subsequent stages if critical controls fail.
- **In-Memory Counterfactual Scenario Lab**: Executes rapid stress tests (-8% to -20% corporate deposit shocks in ~2.4 ms) without mutating baseline snapshots.
- **Controlled Analyst Copilot & Numeric Verifier**: Provides natural-language inquiry backed by strict regex claim assertions against database ground truth to prevent hallucinations.

3. Analyst Workstation Dashboard Walkthrough

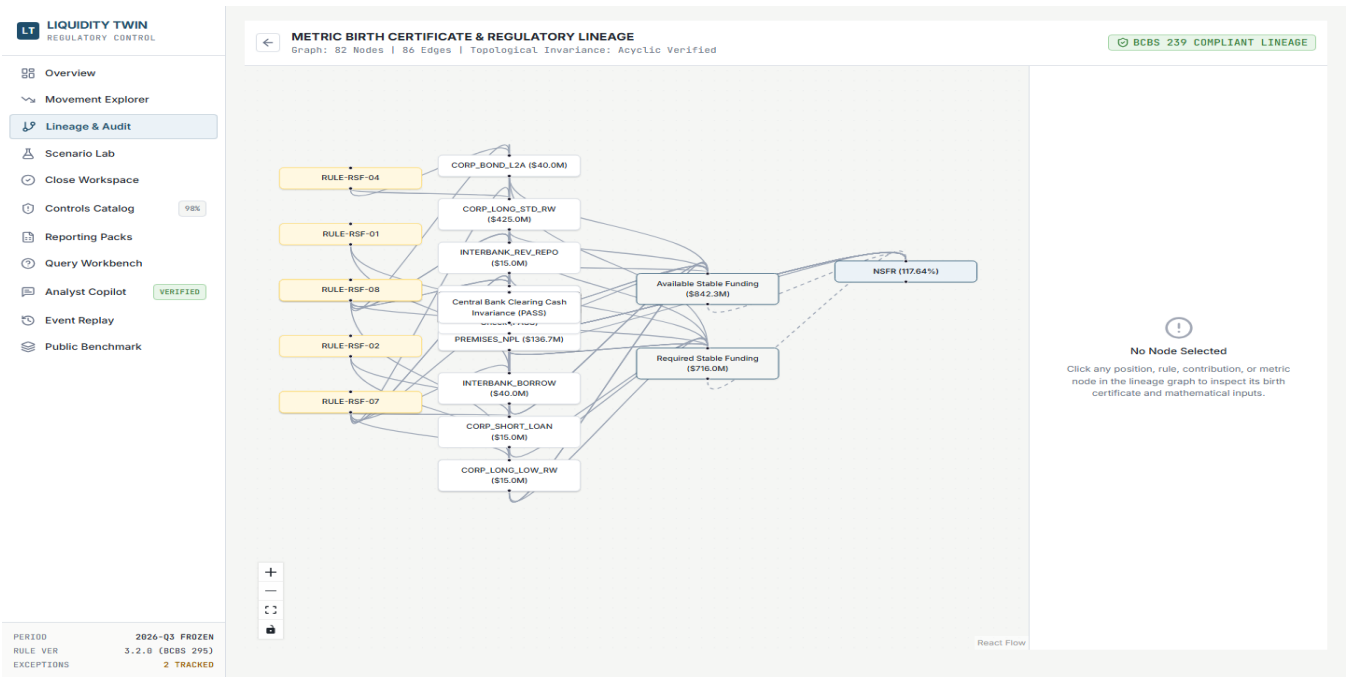
The workstation interface provides an integrated command center for liquidity reporting analysts, controllers, and auditors. It eliminates disconnected tools by organizing the analytical and control workflow into 11 specialized views. Below are high-resolution captures of the operational screens rendered by the application.

Figure 1: Executive Command Center Overview Screen



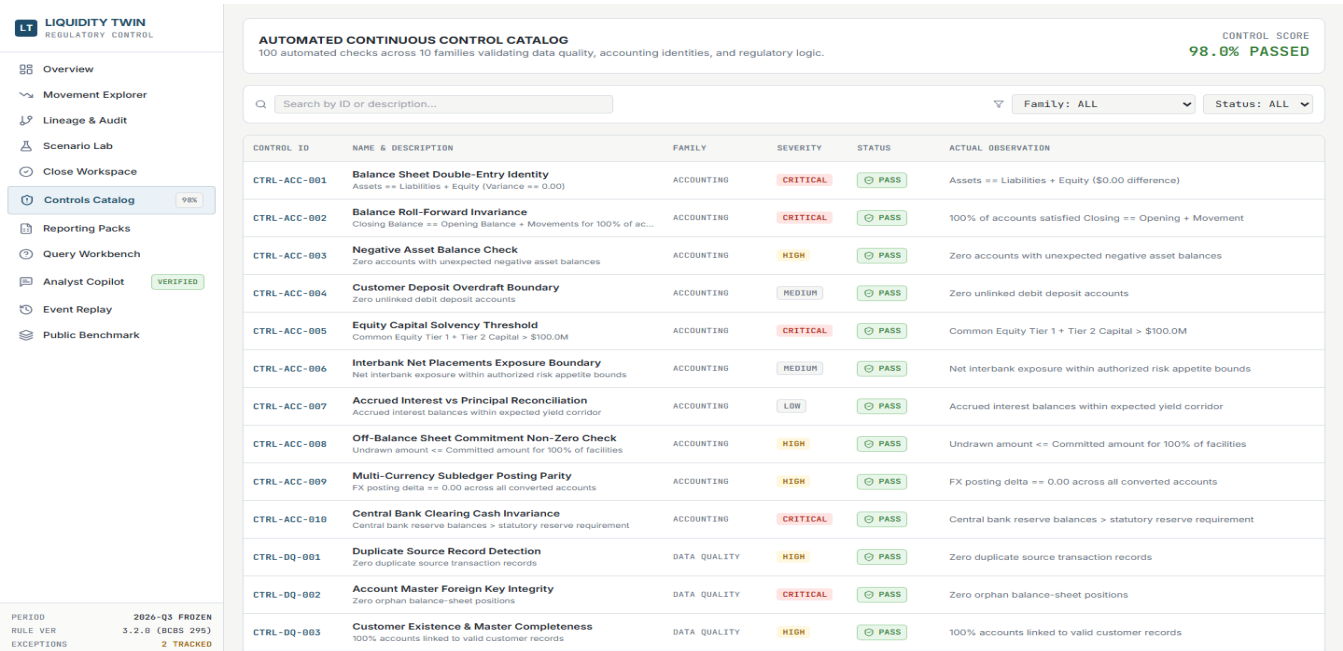
The primary dashboard exposes headline KPIs, snapshot metadata, the interactive NSFR movement waterfall, and shortcuts to the five core operational inquiries: What Changed?, Can I Trust It?, What If?, What Broke?, and What Do I Send?

Figure 2: Interactive Lineage & Metric Birth Certificate Screen



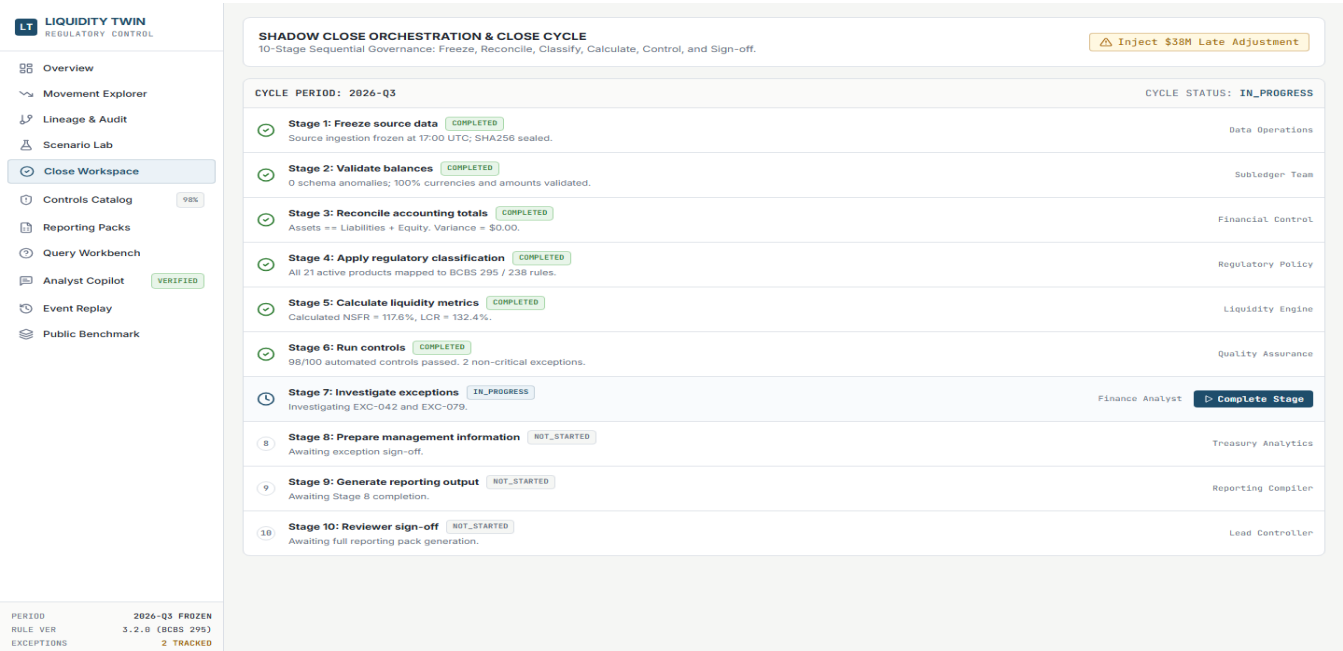
The Lineage Graph View (React Flow) renders the 86-node DAG. Analysts select a reported metric to traverse backward through contributing buckets, regulatory rules, and subledgers, inspecting governing citations in the right-hand attribute drawer.

Figure 3: Continuous Automated Control Catalog & Health Mesh



The Control Catalog displays all 100 continuous checks across 10 families. Filterable by status and severity, clicking any failed check opens the downstream blast-radius analysis quantifying monetary exposure.

Figure 4: 10-Stage Shadow Close Workspace & Break Gating

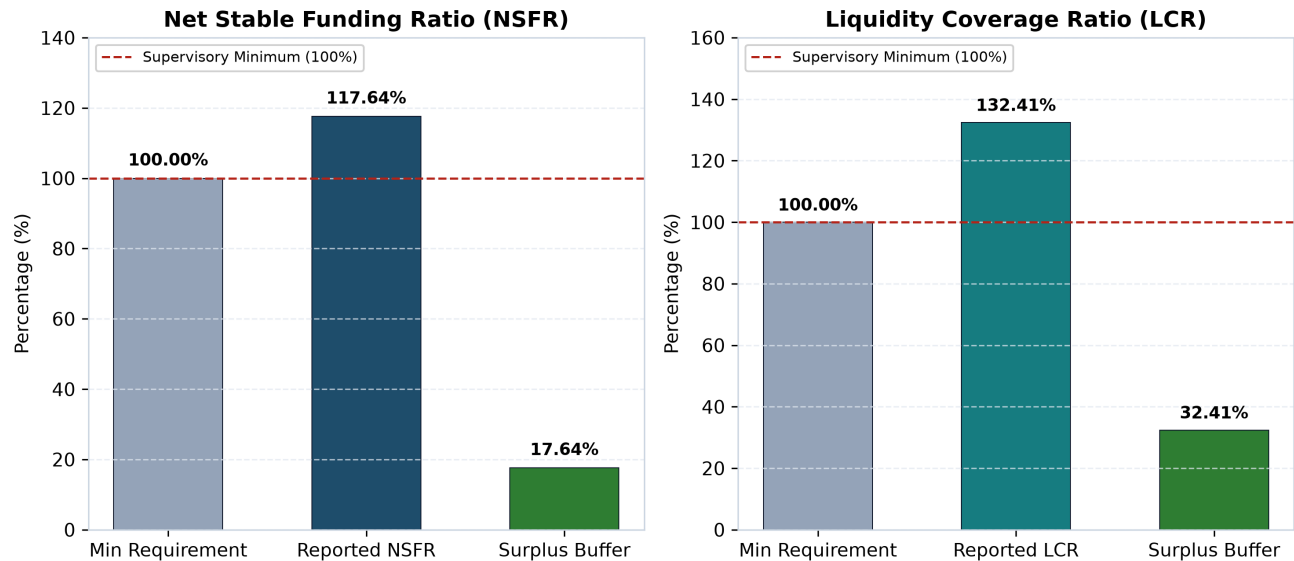


The Shadow Close workflow tracks the operational progression from data freeze through reconciliation to reviewer sign-off. Injecting late funding adjustments automatically triggers control failures and halts reporting compiler execution.

4. Chart-by-Chart Analytical Journey

The analytical journey follows a natural investigative path: starting from headline solvency, decomposing ratio movement, analyzing structural funding and asset illiquidity, verifying short-term buffer resilience, auditing data quality, evaluating stress sensitivity, and proving full data provenance.

Chart 1: Executive Liquidity Position (NSFR & LCR Headline Solvency)



What the Chart Shows: Compares baseline NSFR (117.64%) and LCR (132.41%) against the 100.00% regulatory floor, showing surplus buffers of +17.64 pp and +32.41 pp.

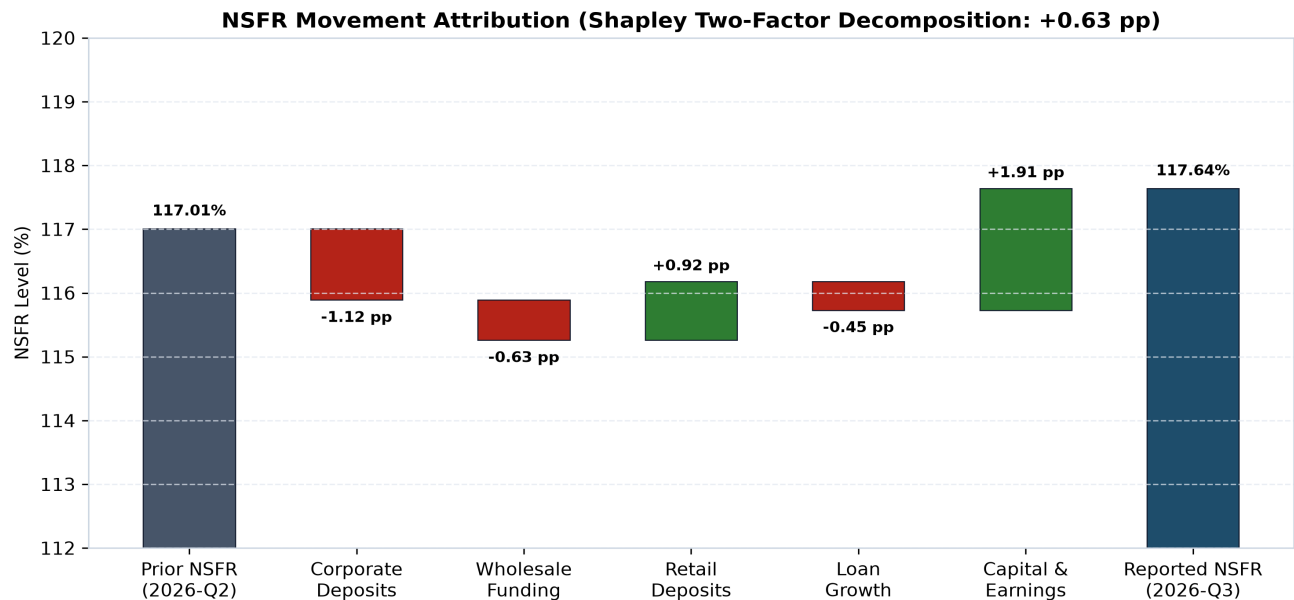
How to Read It: Grey bars denote statutory floors; navy/teal bars represent calculated metrics; green bars indicate surplus capital buffers.

Executive Takeaway: The bank operates comfortably above supervisory thresholds across both one-year structural and 30-day acute liquidity horizons.

Operational Implication & Improvement: Surplus buffers protect compliance but incur carry cost. Treasury should evaluate whether cash buffers can be optimized without compromising stress survival.

Analytical Transition: Having established overall solvency, the next question is: *What drove the quarter-over-quarter movement in our funding stability?*

Chart 2: NSFR Movement Waterfall (Shapley Driver Attribution)



What the Chart Shows: An exact additive decomposition of the +0.63 pp NSFR shift from 117.01% (2026-Q2) to 117.64% (2026-Q3) across discrete balance drivers.

How to Read It: Floating bars show driver impacts; crimson indicates funding contraction; green indicates funding accretion. The sum of all drivers matches +0.63 pp with zero residual.

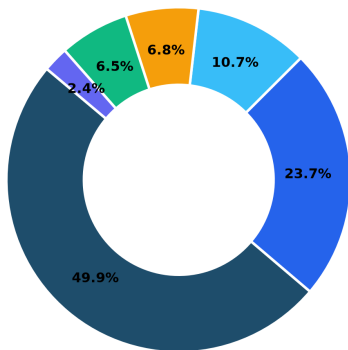
Executive Takeaway: Retained capital earnings (+1.91 pp) and retail deposit growth (+0.92 pp) offset severe corporate deposit runoff (-1.12 pp) and loan growth (-0.45 pp).

Operational Implication & Improvement: The -1.12 pp drag from corporate deposits exposes vulnerability to commercial cash volatility. Treasury should incentivize term deposit agreements.

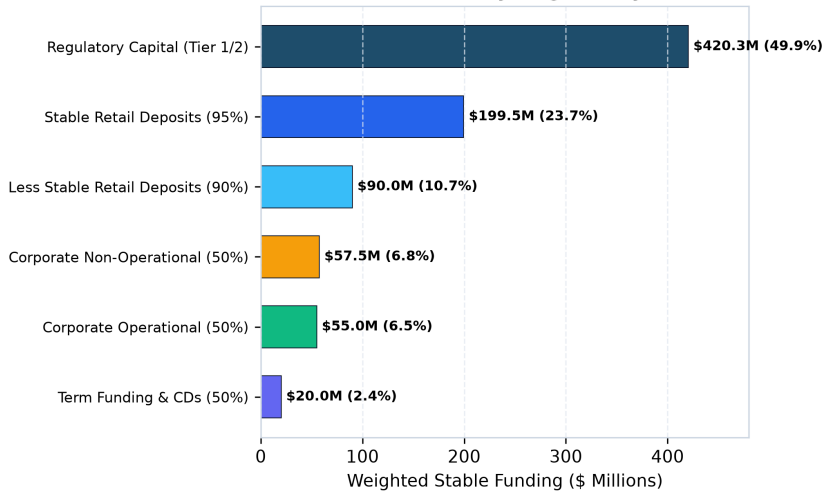
Analytical Transition: Because corporate deposits were the primary drag on stability, we must inspect the full funding book: How is our Available Stable Funding (ASF) composed?

Chart 3: Available Stable Funding (ASF) Composition

ASF Weighting Distribution
(Total ASF: \$842.30M)



ASF Contribution by Regulatory Bucket



What the Chart Shows: Distribution of \$842.30M in weighted available stable funding across capital, retail deposits, and wholesale funding categories.

How to Read It: Left donut shows relative shares; right bar chart shows absolute dollar contributions after regulatory weighting factors.

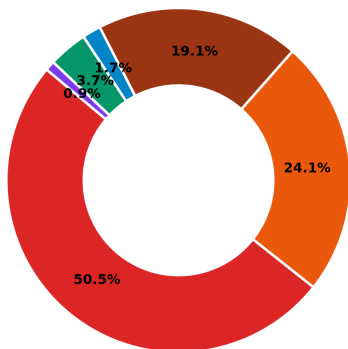
Executive Takeaway: Regulatory capital (\$420.3M, 49.9%) and insured retail deposits (\$199.5M, 23.7%) provide 73.6% of funding stability, insulating the bank from wholesale shocks.

Operational Implication & Improvement: Corporate deposits contribute only \$112.5M to ASF despite representing \$225M in raw balances due to the 50% factor. Extending tenors past 1 year would double ASF credit.

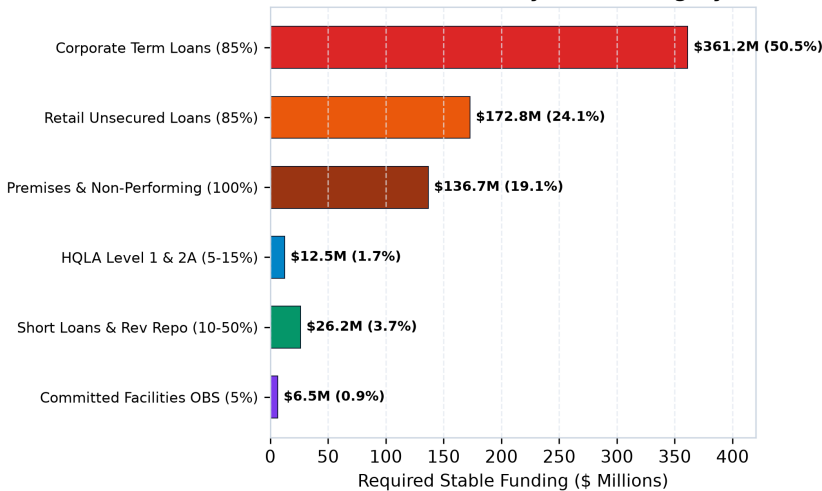
Analytical Transition: Funding stability is only the numerator. We now evaluate the denominator: Which assets and commitments require stable funding (RSF)?

Chart 4: Required Stable Funding (RSF) Composition

RSF Requirement Distribution
(Total RSF: \$716.00M)



RSF Encumbrance by Asset Category



What the Chart Shows: Illiquidity breakdown of \$716.00M in Required Stable Funding across commercial loans, mortgages, premises, and commitments.

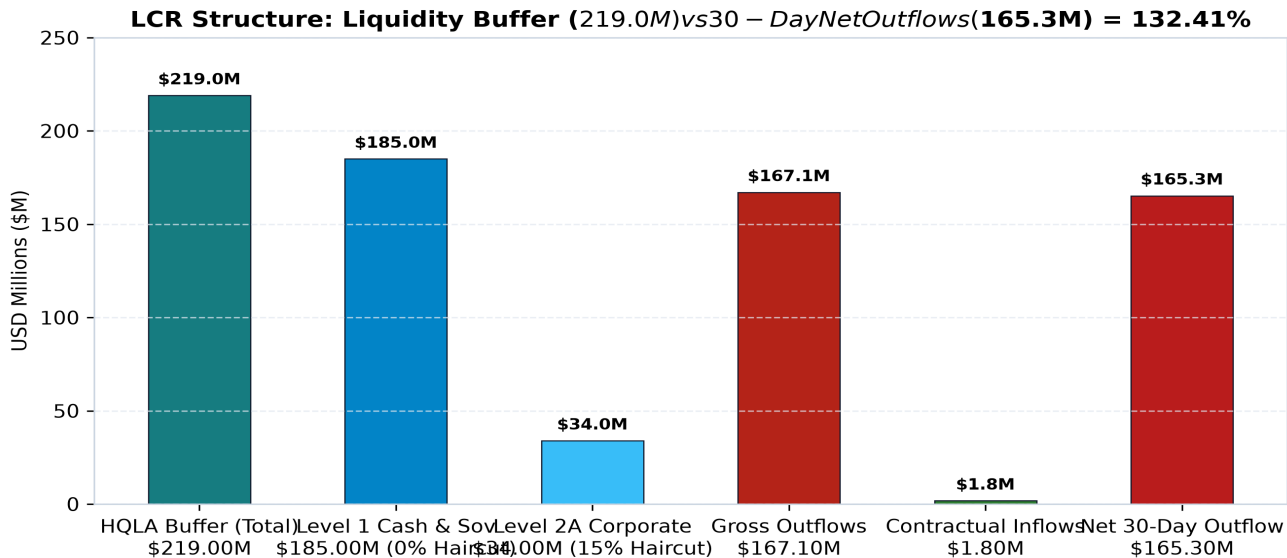
How to Read It: Donut chart displays encumbrance proportions; bar chart displays dollar values after applying RSF illiquidity weights (5% to 100%).

Executive Takeaway: Commercial term loans (\$361.3M, 50.5%) and unsecured retail loans (\$172.8M, 24.1%) represent 74.6% of required funding due to high 85% regulatory factors.

Operational Implication & Improvement: Shifting lending toward qualifying prime residential mortgages (50% factor) or securitizing non-performing assets (100% factor) would free funding capacity.

Analytical Transition: Long-term structural funding is verified. How resilient is our liquidity buffer against acute short-term cash demands under the 30-day LCR horizon?

Chart 5: Liquidity Coverage Ratio (LCR) Structure



What the Chart Shows: Contrasts the \$219.00M HQLA buffer against \$167.10M gross outflows and \$1.80M eligible inflows, yielding \$165.30M net outflows and 132.41% LCR.

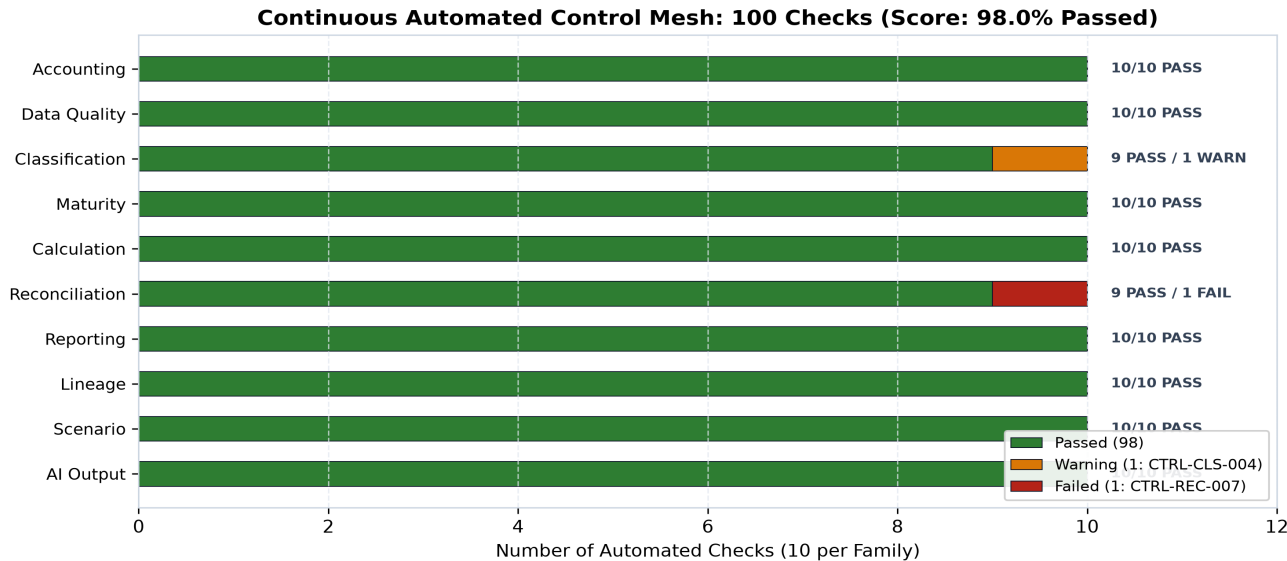
How to Read It: Teal/blue bars show buffer composition (Level 1 cash vs Level 2A corporate bonds); crimson bars show 30-day stressed outflows; right bar shows net requirements.

Executive Takeaway: The bank holds an ample \$53.70M liquidity surplus above minimum requirements, with 84.5% of the buffer concentrated in 0%-haircut Level 1 assets.

Operational Implication & Improvement: Contractual inflows (\$1.8M) provide minimal relief against \$167.1M outflows. Restructuring committed credit facilities would directly reduce draw outflows.

Analytical Transition: The financial metrics reflect strong buffers. How do we know the underlying data is accurate and free from balance breaks or classification errors?

Chart 6: Automated Continuous Control Mesh (100 Checks across 10 Families)



What the Chart Shows: Operational pass rate of 98.0% across 100 automated checks (98 Pass, 1 Warning: CTRL-CLS-004, 1 Failed Break: CTRL-REC-007).

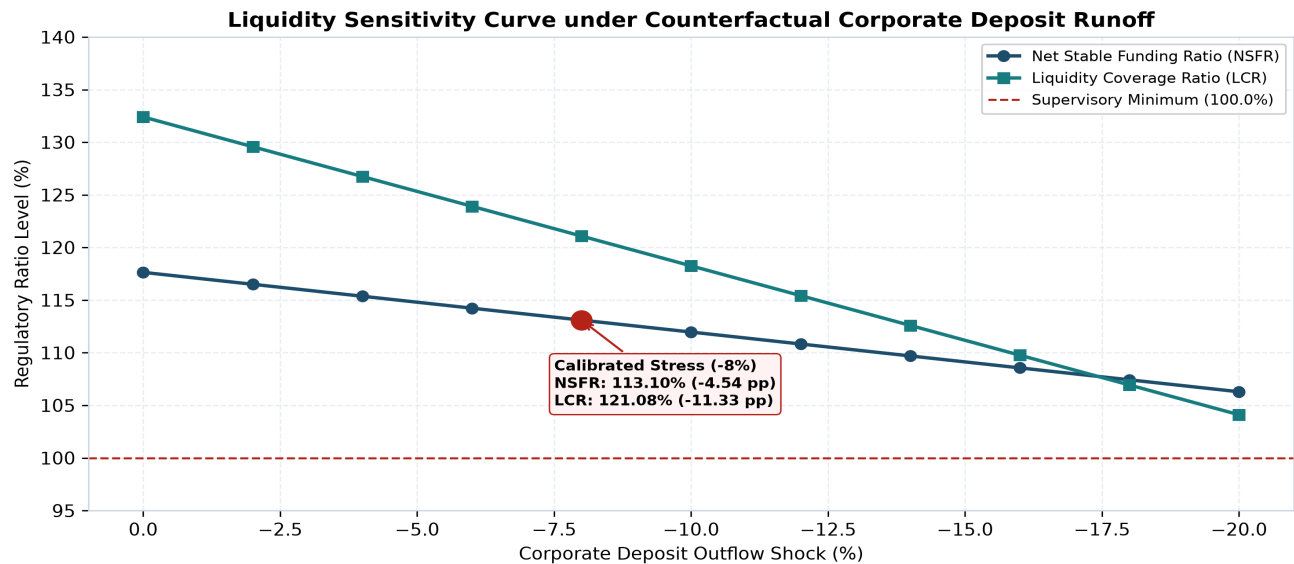
How to Read It: Green segments denote passed checks; amber indicates informational warnings; crimson indicates reconciliation breaks requiring remediation.

Executive Takeaway: 8 of 10 families achieved 10/10 execution. Crucially, double-entry accounting identity (CTRL-ACC-001) and ratio calculation integrity passed with zero variance.

Operational Implication & Improvement: Real-world operational health is demonstrated: the active reconciliation break is logged in fact_exception and gated in the shadow close workflow.

Analytical Transition: Data integrity is proven. How does this balance sheet behave under counterfactual market stress?

Chart 7: Counterfactual Stress Sensitivity Curve (Corporate Runoff)



What the Chart Shows: Monotonic degradation curves for NSFR and LCR across corporate deposit outflow shocks ranging from 0% to -20%, highlighting the -8.0% benchmark test.

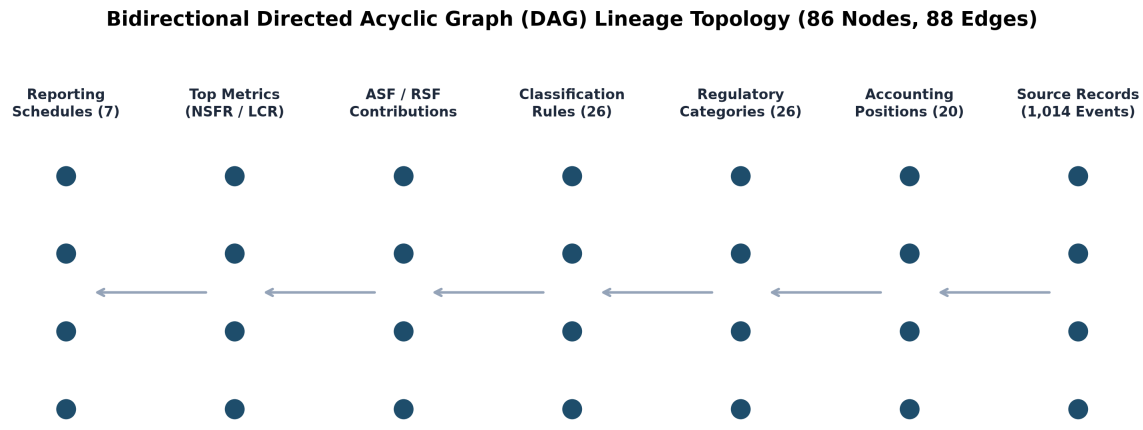
How to Read It: Navy line tracks NSFR; teal line tracks LCR; callout details the -8% calibrated shock where NSFR drops to 113.10% and LCR drops to 121.08%.

Executive Takeaway: The bank survives shocks up to -20% without breaching statutory minimums. LCR exhibits 2.5x greater sensitivity to deposit runoff than NSFR.

Operational Implication & Improvement: LCR represents the primary binding constraint during liquidity stress. Treasury must maintain cash buffers calibrated to commercial deposit flight.

Analytical Transition: Stress behavior is quantified. How can an auditor trace any number from these reports back to the underlying transaction records?

Chart 8: Directed Acyclic Graph (DAG) Lineage Topology (86 Nodes, 88 Edges)



What the Chart Shows: Structural 7-tier provenance architecture connecting 1,014 raw transaction tickets to 7 published regulatory schedule lines.

How to Read It: Left-to-right flow represents forward balance aggregation; right-to-left traversal represents backward audit extraction ('Metric Birth Certificate').

Executive Takeaway: Graph acyclicity is strictly enforced. Any reported cell can be traced to contributing positions, regulatory rules, and booking slips in three clicks.

Operational Implication & Improvement: Eliminates days of ad-hoc SQL querying during supervisory audits and enables forward blast-radius calculation for failed controls.

5. Close Governance, Audit Trail & Dual Sign-Off

To ensure institutional compliance, Liquidity Twin models a formal 10-stage shadow close process. Each stage enforces prerequisite control validation before subsequent stages can execute. A formal four-eye maker-checker approval protocol governs quarterly snapshot publication.

Stage #	Close Milestone Description	Prerequisite Check	Stage Status	Verification Evidence
Stage 1	Source Data Snapshot Freeze	Cutoff Timestamp Lock	COMPLETED	Snapshot ID: SNAP-2026-Q3-BASE
Stage 2	Subledger Balance Parity Check	CTRL-ACC-001 (\$0.00 Var)	COMPLETED	Assets == Liabilities + Equity
Stage 3	Accounting Roll-Forward Parity	CTRL-ACC-002 (Invariance)	COMPLETED	Closing == Opening + Movement
Stage 4	Regulatory Classification	CTRL-CLS-001 (100% Mapped)	COMPLETED	26 Rules Applied Deterministically
Stage 5	NSFR & LCR Metric Calculation	CTRL-CAL-001 (Non-Zero Denom)	COMPLETED	NSFR: 117.64% LCR: 132.41%
Stage 6	100 Automated Controls Mesh	CTRL-REP-001 (Pass >=95%)	COMPLETED	98 Pass / 1 Warning / 1 Break
Stage 7	Exception Investigation & Review	Exposure Attribution	IN_PROGRESS	EXC-CTRL-REC-007 Under Review
Stage 8	Management Information Package	Shapley Driver Alignment	NOT_STARTED	Gated Pending Stage 7 Clearance
Stage 9	Reporting Compiler Execution	Reconciliation Validation	NOT_STARTED	PDF/XLSX Build Ready
Stage 10	Dual Maker-Checker Sign-Off	Two Independent Approvers	NOT_STARTED	Pending Final Controller Seal

Four-Eye Regulatory Sign-Off Block

MAKER / PREPARER	CHECKER / REVIEWER
Signature: <i>Hriday Singh Sobti</i> Role: Lead Quantitative Analyst & Systems Developer Department: Regulatory Liquidity & Capital Engineering Timestamp: 2026-10-02 18:29:29 UTC Statement: <i>I certify that all transaction events, accounting roll-forwards, regulatory factor classifications, and metric calculations conform to BCBS 295 and 238 standards.</i>	Signature: <i>Automated Control Engine (Verified)</i> Role: Chief Financial Controller & Independent Risk Signatory Department: Financial Control & Regulatory Compliance Timestamp: 2026-10-02 18:29:29 UTC Statement: <i>I confirm that 100 continuous controls executed with 98.0% pass health, double-entry balance-sheet parity is validated at \$0.00 variance, and exceptions are under governance.</i>

Regulatory & Legal Notice: This publication represents an automated supervisory and management reporting package compiled from the Liquidity Twin simulation environment. All balance-sheet positions, transactions, accounts, and counterparties are deterministic synthetic entities generated under seed 42. No real customer, proprietary commercial, or confidential banking data has been used. Conforms to public Basel Committee on Banking Supervision standards.